

Jashore University of Science and Technology

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PROJECT REPORT ON

Course Title: Electronics-I Sessional

Course Code: EEE-2102

Project Name: Short circuit and Overload protection System.

Submitted By

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Aim:

- To design a protection system for short circuits and overloads.
- To ensure the safety of electrical appliances and circuits.
- To provide an indication (using LEDs) of normal and fault conditions.

Introduction:

Short circuits are a common issue in electrical and electronic circuits, often leading to damage to components, overheating, or even fire hazards. This project aims to design a simple and effective short circuit protection system using a relay, transistor, and a 100k variable resistor. The system detects excessive current flow due to a short circuit and disconnects the load to prevent damage.

Apparatus required:

S.L no	Apparatus Name	Ratings	Quantity
1	Transistor (BC547)	45 V ,100mA	1
2	Variable resistor	100K Ω	1
3	Relay	12V	1
4	Led	2 V, 20mA	2
5	Push switch	-	1
6	On/Off switch	-	1
7	Bazar	9V	1
8	Terminal block	-	1
9	Power source	12 V	1

Circuit Diagram:

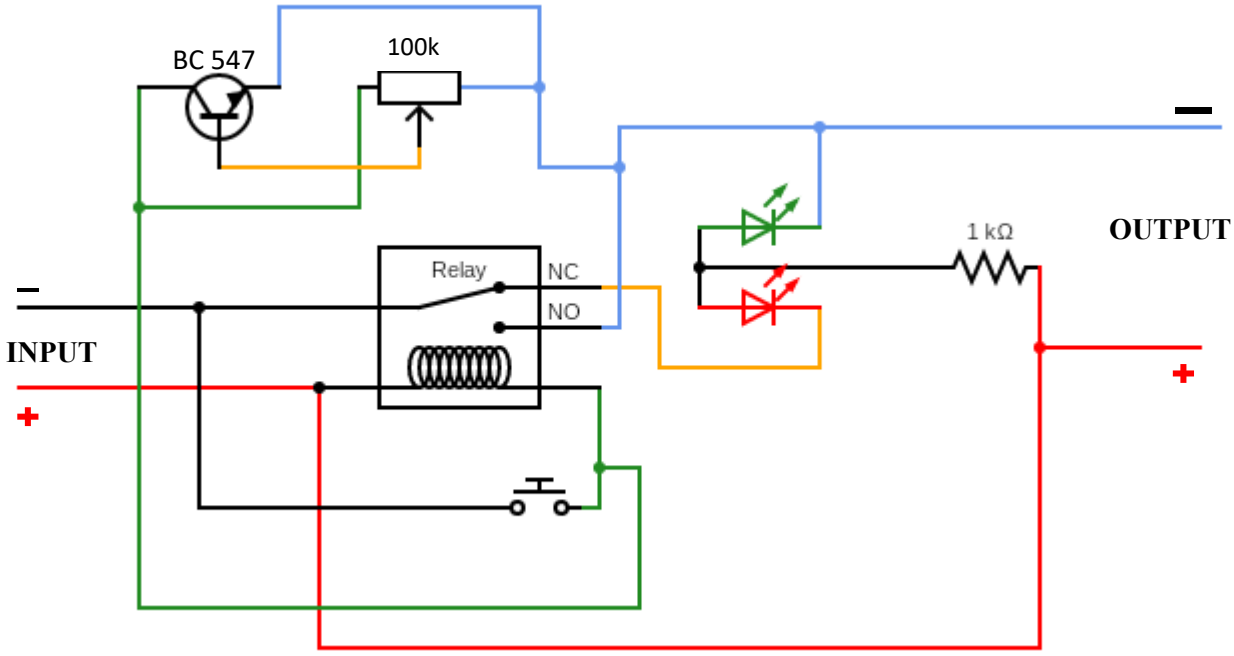


Figure 1: Short circuit and Overload protection system circuit

Circuit Description:

- Transistor (BC547):
  - ❖ Acts as a switch to control the relay coil.
  - ❖ Detects overcurrent conditions and interrupts the circuit to protect the load.
  - ❖ May amplify control signals to ensure reliable relay operation.
- Potentiometer:
  - ❖ Adjusts the current threshold for overload protection.
  - ❖ Provides fine-tuning of the circuit's sensitivity to overcurrent conditions.
  - ❖ May be part of a voltage divider or feedback mechanism for control purposes.
- Resistors: For current limiting to the LED
- LED: Indicates the status of the circuit (optional).
- Power Supply: DC power source (12V).
- Switch: Push switch use for manual reset and other use for on/off the system.
- Terminal Block: connect the power supply and load.
- Bazar: Alarm the short circuit and overload condition.

Working principle:

- Normal Operation:
  - ❖ When the circuit is powered and operating within normal current limits, the transistor (BCS47) is in the "ON" state, allowing current to flow from the input to the output, thus powering the load, indicating the red LED.
  - ❖ The 1K resistor helps maintain a stable current flow by providing a specific resistance that limits the current to a safe level.

- Overload Condition:
  - ❖ If the current drawn by the load exceeds the safe threshold, the voltage drop across the 1K resistor increases.
  - ❖ This increased voltage drop can be sensed by the transistor.
  - ❖ When the sensed voltage exceeds a certain level, the transistor turns off, interrupting the current flow to the load, this de-energizes the relay coil, causing the relay to switch to its default state (NC) and cutting off power to the load.

- Short Circuit Condition:
  - ❖ In the event of a short circuit, the current would spike dramatically, causing a significant voltage drop across the 1K resistor.
  - ❖ The protection circuit detects this excessive current and quickly shuts off the transistor, effectively disconnecting the load from the power supply.
  - ❖ This de-energizes the relay coil, causing the relay to switch to its default state (NC) and cutting off power to the load.

- Reset Function:
    - ❖ After an overload or short circuit event, the circuit may remain in a protected state (transistor off) to prevent further damage.
    - ❖ The "RESET" button allows the user to manually reset the circuit once the fault condition has been resolved. Pressing the reset button restores the circuit to its normal operating state, allowing current to flow again.

Advantages:

- Simple and cost-effective circuit.
- Adjustable sensitivity using the 100k potentiometer.
- Provides automatic disconnection in case of short circuits and overload.
- Can be used in various applications such as battery protection, motor protection, and general circuit safety.

Applications:

- Power supply protection.
- Battery management systems.

Discussion:

In this project we try our best to fulfill our project. From this project we ready a system that prevent electronics component for damaging because of short circuit and overload. We also include in this system in manual reset. The manual reset function helps to manually reset the circuit, returning the relay to its default NC (Normally Closed) position and restoring power to the load. Also feature of this project to set threshold current limit use potentiometer. However, at beginning of the prepare project we face many problems when we use mosfet instead of relay for switching. We could not switch mosfet perfectly. This problem occur may be proper biasing. After we use relay and successful our aim. Thre have a problem in our project, when we supply a voltage the circuit work in fault condition, because fault condition set in NC contact in relay. At first relay coil not energized.